

DMIMS Full Database Schema & Migration Guide

Supporting Document 1

1. Purpose

This document guides the developer to create the full DMIMS database structure in the correct order.

DMIMS uses:

Laravel migrations

MySQL / MariaDB

customer_id-based company isolation

soft deletes for master records

immutable movement/audit logs

2. Database Design Rules

2.1 customer_id Rule

Every customer-owned table must include:

customer_id

This applies to:

users

categories

products
locations
product_location_stocks
stock_movements
stock_alerts
boxes
document_files
document_movement_logs
barcode_registry
barcode_scan_logs
customer_subscriptions
subscription_logs
licenses
license_logs
billing_records
billing_payments
billing_logs
audit_logs
notifications

2.2 Platform Tables Without customer_id

These tables are platform-wide:

roles
permissions
modules

subscription_plans

location_types

document_types if global

settings if global

Some tables may allow `customer_id = NULL` for global/default records.

2.3 Soft Delete Rule

Use soft deletes for master data:

customers

users

categories

products

locations

boxes

document_files

barcode_registry

billing_records

Do not soft delete:

stock_movements

document_movement_logs

audit_logs

subscription_logs

license_logs

billing_logs

barcode_scan_logs

These are history logs and must be immutable.

2.4 Index Rule

Add indexes for:

customer_id

status

barcode

product_id

location_id

box_id

movement_type

action_type

performed_at

created_at

valid_to

expected_return_date

source_type

destination_type

payment_status

billing_status

license_status

subscription_status

3. Recommended Migration Order

Create migrations in this order:

01 customers

02 users update customer fields

03 modules

04 customer_modules

05 subscription_plans

06 customer_subscriptions

07 subscription_logs

08 licenses

09 license_logs

10 billing_records

11 billing_payments

12 billing_logs

13 location_types

14 locations

15 categories

16 products

17 product_location_stocks

18 stock_movements

19 stock_alerts

20 document_types

21 boxes

22 document_files

23 document_movement_logs

24 barcode_registry

25 barcode_scan_logs

26 settings

27 audit_logs

28 notifications

Why this order matters:

Customers must exist before company-owned records.

Locations must exist before products/boxes can use them.

Boxes must exist before files can be assigned.

Subscription/license/billing should exist before access control is enforced.

Movement logs come after their parent records.

4. Core Tables

4.1 customers

Purpose:

Stores customer company information.

Required fields:

id

company_name

company_code

registration_no

tin_no

contact_person

email

phone

address

status

notes

created_by

updated_by

created_at

updated_at

deleted_at

Important rules:

company_code must be unique.

Use soft delete.

Do not hard delete active customer records.

4.2 users

Purpose:

Stores platform and company users.

Add these fields to Laravel users table:

customer_id nullable

username nullable

employee_id nullable

phone nullable

job_title nullable

status

is_platform_user boolean

last_login_at nullable

created_by nullable

updated_by nullable

deleted_at nullable

Rules:

Datamation internal user: customer_id = NULL, is_platform_user = true.

Company user: customer_id = customer id, is_platform_user = false.

Email must be unique.

4.3 modules

Purpose:

Stores available system modules.

Fields:

id

module_code

module_name

description

status

created_at

updated_at

Default modules:

stock_inventory
document_tracking
barcode_scanning
barcode_printing
reports
import_export
advanced_audit
backup_restore
billing_view

4.4 customer_modules

Purpose:

Controls which modules are enabled for each customer.

Fields:

id

customer_id

module_id

is_enabled

enabled_at

disabled_at

created_by

updated_by

created_at

updated_at

Unique rule:

customer_id + module_id must be unique.

5. Subscription Tables

5.1 subscription_plans

Purpose:

Stores reusable subscription plans.

Fields:

id

plan_code

plan_name

description

max_users

max_products

max_document_files

max_boxes

enabled_modules JSON

allowed_reports JSON

billing_cycle

price

status

created_at

updated_at

Billing cycles:

monthly

quarterly

half_yearly

yearly

custom

Rules:

Plan code must be unique.

Inactive plans cannot be assigned to new customers.

5.2 customer_subscriptions

Purpose:

Stores actual subscription assigned to a customer.

Fields:

id

customer_id

subscription_plan_id

subscription_no

valid_from

valid_to

grace_period_days

max_users

max_products

max_document_files

max_boxes

enabled_modules JSON

allowed_reports JSON

billing_cycle

status

renewal_notes

created_by

updated_by

created_at

updated_at

Statuses:

trial

active

near_expiry

expired_grace

expired

cancelled

Rules:

Subscription controls plan, modules, limits, dates and billing cycle.

It does not decide final technical access.

Final access is controlled by license.

5.3 subscription_logs

Purpose:

Tracks subscription changes.

Fields:

id

customer_id

customer_subscription_id

action

old_value JSON

new_value JSON

remarks

performed_by

performed_at

created_at

updated_at

Actions:

created

renewed

plan_changed

limits_changed

modules_changed

expired

cancelled

6. License Tables

6.1 licenses

Purpose:

Controls final technical access.

Fields:

id

customer_id

license_no

license_status

technical_access_mode

valid_from

valid_to

grace_period_days

remarks

created_by

updated_by

created_at

updated_at

License statuses:

active

suspended

revoked

expired

Technical access modes:

full_access

view_only

blocked

Mapping:

active = full_access

suspended = view_only

expired = view_only

revoked = blocked

Rules:

Each customer should have one current license.

License decides final technical permission.

6.2 license_logs

Purpose:

Tracks license changes.

Fields:

id

customer_id

license_id

action

old_value JSON

new_value JSON

remarks

performed_by

performed_at

created_at

updated_at

Actions:

created

renewed

suspended

revoked

expired

reactivated

7. Billing and Manual Payment Tables

7.1 billing_records

Purpose:

Stores billing/invoice records.

Fields:

id

customer_id

customer_subscription_id

license_id
billing_no
invoice_no
billing_period_from
billing_period_to
invoice_date
due_date
currency
subtotal_amount
tax_amount
discount_amount
total_amount
paid_amount
balance_amount
billing_status
payment_status
remarks
created_by
updated_by
created_at
updated_at
deleted_at

Billing statuses:

draft
issued

sent

partially_paid

paid

overdue

cancelled

void

Payment statuses:

unpaid

paid

partially_paid

overdue

waived

cancelled

refunded

Rules:

Payment is updated manually by Datamation Super Admin.

No online payment portal is required.

Customer users cannot edit billing/payment records.

7.2 billing_payments

Purpose:

Stores manual payment entries.

Fields:

id
billing_record_id
customer_id
payment_date
payment_amount
payment_method
payment_reference_no
payment_status
proof_file_path
remarks
received_by
created_at
updated_at

Payment methods:

Bank Transfer

Cash

Cheque

Online Transfer

Internal Adjustment

Waived

Other

7.3 billing_logs

Purpose:

Tracks billing and payment changes.

Fields:

id

customer_id

billing_record_id

action

old_value JSON

new_value JSON

remarks

performed_by

performed_at

created_at

updated_at

8. Location and Stock Tables

8.1 location_types

Purpose:

Defines location type master.

Examples:

Warehouse

Building

Floor

Room

Rack

Shelf

Cabinet

Fields:

id

type_code

type_name

description

sort_order

status

created_at

updated_at

8.2 locations

Purpose:

Stores physical locations inside DMIMS control.

Fields:

id

customer_id

parent_id

location_type_id

location_code

location_name

full_path

barcode

can_store_stock

can_store_boxes

status

created_by

updated_by

created_at

updated_at

deleted_at

Rules:

Locations are shared by stock and document modules.

Do not create fake locations like CUSTOMER, LAWYER, OUTSIDE.

External source/destination must be stored in movement fields.

8.3 categories

Purpose:

Stores product categories.

Fields:

id

customer_id

category_code

category_name

description

status

created_by

updated_by

created_at

updated_at

deleted_at

8.4 products

Purpose:

Stores stock products.

Fields:

id

customer_id

category_id

default_location_id

sku

barcode

product_name

description

reorder_level

unit_cost

unit_price

status

created_by

updated_by

created_at

updated_at

deleted_at

Unique:

customer_id + sku

customer_id + barcode

8.5 product_location_stocks

Purpose:

Stores stock balance per product per location.

Fields:

id

customer_id

product_id

location_id

quantity_on_hand

reserved_quantity

available_quantity

last_movement_at

created_at

updated_at

Unique:

customer_id + product_id + location_id

8.6 stock_movements

Purpose:

Stores immutable stock movement history.

Fields:

id

customer_id

movement_no

product_id

movement_type

from_location_id

to_location_id

quantity

unit_cost

reference_no

reason

remarks

source_type

source_name

source_reference_no

destination_type

destination_name

destination_reference_no

performed_by

performed_at

created_at

updated_at

Movement types:

opening_balance

receive_in

stock_out

internal_transfer

adjustment

return

disposal

Rules:

Do not delete movement logs.

Use database transactions for all stock movements.

8.7 stock_alerts

Purpose:

Stores low stock and out-of-stock alerts.

Fields:

id

customer_id

product_id

location_id

alert_type

threshold_quantity

current_quantity

status

created_at

updated_at

Alert types:

low_stock

out_of_stock

overstock

9. Document Tracking Tables

9.1 document_types

Purpose:

Stores document type master.

Fields:

id

customer_id nullable

type_code

type_name

description

status

created_at

updated_at

Rule:

customer_id NULL = global document type.

customer_id filled = customer-specific document type.

9.2 boxes

Purpose:

Stores physical archive/document boxes.

Fields:

id

customer_id

current_location_id

box_barcode

box_number

source_origin

capacity_limit

current_file_count

status

expected_return_date

actual_return_date

last_internal_location_id

remarks

created_by

updated_by

created_at

updated_at

deleted_at

Statuses:

active

closed

moved_out

archived

damaged

missing

Rules:

Do not update every file when box moves.

File location is derived from box location.

9.3 document_files

Purpose:

Stores physical document files.

Fields:

id

customer_id

current_box_id

document_type_id

file_barcode

file_reference_no

title

owner_name

current_status

source_origin

destination

expected_return_date

actual_return_date

last_internal_box_id

received_date

archived_date

remarks

created_by

updated_by

created_at

updated_at

deleted_at

Statuses:

active

transferred

moved_out

archived

missing

damaged

closed

9.4 document_movement_logs

Purpose:

Stores immutable file/box movement history.

Fields:

id

customer_id

movement_no

movable_type

movable_id

action_type

from_location_id

to_location_id

from_box_id

to_box_id

source_type

source_name

source_reference_no

destination_type

destination_name

destination_reference_no

external_contact_person

external_contact_phone

external_contact_email

external_address

expected_return_date

handover_reference_no

received_by

returned_by

scanned_barcode

remarks

performed_by

performed_at

created_at

updated_at

Movable types:

document_file

box

Action types:

receive_in_file

receive_in_box

transfer_file

transfer_box

move_out_file

move_out_box

return_file

return_box

archive

correction

Rules:

Do not edit/delete movement logs.

Wrong movements must be corrected using correction action.

10. Barcode Tables

10.1 barcode_registry

Purpose:

Central barcode lookup table.

Fields:

id

customer_id

barcode

barcode_type

reference_table

reference_id

status

printed_count

last_scanned_at

created_by

updated_by

created_at

updated_at

deleted_at

Barcode types:

product

location

box

document_file

Unique:

customer_id + barcode

10.2 barcode_scan_logs

Purpose:

Stores scan history.

Fields:

id

customer_id

barcode

barcode_type

reference_table

reference_id

scan_result

action_taken

scanned_by

scanned_at

ip_address

user_agent

created_at

updated_at

Scan results:

found

unknown

inactive

permission_denied

11. System Tables

11.1 settings

Purpose:

Stores platform and customer settings.

Fields:

id

customer_id nullable

setting_group

setting_key

setting_value

setting_type

created_at

updated_at

Rule:

customer_id NULL = global setting.

customer_id filled = customer-specific setting.

11.2 audit_logs

Purpose:

Stores system audit trail.

Fields:

id

customer_id nullable

user_id nullable

module

action

auditable_type

auditable_id

old_values JSON

new_values JSON

ip_address

user_agent

remarks

created_at

updated_at

11.3 notifications

Purpose:

Stores user/system notifications.

Fields:

id

customer_id nullable

user_id nullable

notification_type

title

message

is_read

read_at

created_at

updated_at

12. Seed Data Required

Create seeders for:

Roles

Permissions

Modules

Default subscription plans

Location types

Default document types

Datamation Super Admin user

Datamation Management user

13. Migration Testing Checklist

After migrations, verify:

`php artisan migrate:fresh --seed` works

All tables are created

Foreign keys are valid

Indexes are created

Super Admin user exists

Management user exists

Default modules exist

Default plans exist

No migration error occurs

Company isolation fields exist

Soft deletes are present where required

Movement logs are immutable

14. Final Developer Rule

The database must support:

Multi-company customer isolation

Subscription control

License control

Manual billing/payment

Stock inventory

Document tracking

Internal transfer

External receive-in

External move-out

Barcode registry

PWA-ready system

Reports and analytics

Full audit trail